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TECHNICAL PAPER

# Design and fabrication of leaf-based microcantilever beams

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**Abstract**

Microelectromechanical Systems (MEMS) cantilever beams, known for their versatility and wide-ranging applications, are extensively used in various engineering fields. In this study, we examine the frequency and damping characteristics of MEMS cantilever beams with different shapes, including uniform, diamond, hexagonal, and leaf-shaped configurations derived from a stepped rectangular cantilever beam by sweeping the vertical faces. We determined the frequency and damping characteristics of these shapes through numerical simulations. Additionally, we conducted experiments on the leaf-shaped beams. The numerical and experimental frequency values showed good agreement, with a maximum percentage error of 11.2%. Finally, we demonstrated the use of differently shaped beams as suspension springs in MEMS accelerometers. Our findings indicate that the unique geometry of leaf-shaped beams allows for the effective separation of different vibration modes:in-plane, torsional, and out-of-plane. This mode separation provides a distinct advantage over other beam shapes, such as square or hexagonal beams, by efficiently improving frequency tuning and isolating different vibrational modes.

## 1 Introduction

Microelectromechanical Systems (MEMS) cantilever-based microbeams are pivotal in various applications ranging from sensing and actuation to biosensing, gas sensing, and energy harvesting (Jianxiong et al. 2019). These microbeams leverage the mechanical vibrations of micro-scale structures to achieve high sensitivity and precision, making them indispensable components in modern technology (Hossain et al. 2024). One of the key factors influencing the performance of MEMS cantilever resonators is their shape, which dictates their mechanical properties and resonant behavior (Hocheng et al. 2012). In recent years, researchers have explored many cantilever shapes, ranging from conventional rectangular geometry (Peng et al. 2024; Fernando et al. 2007) to more complex configurations such as arrow (Ashok et al. 2019), trapezoidal (Hosseini and Hamedi 2016; Ashok et al. 2018; Zhang et al. 2017; He et al. 2020; Ansari et al. 2009), V-shape (Subramanian and Gupta 2009; Dewanjee et al. 2024; Damircheli and Eslami 2020), hexagonal (Gangele et al. 2019; Liao et al. 2022; Wang et al. 2023), curved (Krylov et al. 2011; Ouakad and Younis 2014; Jujjuvarapu and Pandey 2024), converging and diverging beams (Devsoth and Pandey 2024; Singh and Pandey 2015). Each of these shapes offers unique advantages and characteristics, leading to diverse applications and performance enhancements.

In addition to these considerations, numerous investigations have focused on optimizing micro-cantilever beam shapes to improve device performance. This optimization involves adjusting critical parameters like length, width, thickness, and cross-sectional shape to fine-tune stiffness, mass distribution, natural frequencies, and damping characteristics. Pradeesh et al. (2019) investigated the performance of various designs of cantilever-based piezoelectric energy harvesters with multiple materials, comparing them

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with single-piezoelectric-mounted energy harvesters using experimental and numerical methods, finding that certain designs yield significantly higher power output and lower resonant frequencies than conventional rectangular harvesters. Rami Reddy et al. (2016) utilized a cantilever-based piezoelectric energy harvester with a cavity, demonstrating that the cavity increases strain and voltage generation, yielding a 75% higher voltage than a beam without a cavity. Sahoo and Pandey (2018) explored the impact of non-uniform beams, specifically those with linear and quartic width variations, on the performance of cantilever-based piezoelectric energy harvesters (PEHs), revealing that a diverging beam with a tapering parameter of 0.6 can achieve a 3833% increase in power output compared to a uniform beam, thus suggesting potential for wide-band PEH design incorporating arrays of uniform and non-uniform beams. Wang et al. (2024) presented a novel design and fabrication method for a hollow structure cantilever probe micro force sensor using femtosecond laser-induced two-photon polymerization 3D microprinting technology, offering tunable elastic constants and high force sensitivity, with potential applications in biomechanical detection due to its excellent stability, reproducibility, and flexibility. Upadhyaya et al. (2021) analyzed the performance of different microcantilever shapes integrated with an optical MEMS system, highlighting the novel concept of coupling optical sensing layers with various shapes such as rectangular, trapezoidal, and triangular profiles, with notable results indicating the superior sensitivity and Q factor of the rectangular microcantilever shape, particularly in air mediums, suggesting significant potential for sensing biofluids and device miniaturization. Ahsan and Aurelil (2018) explore the use of shape-morphing deformation in a submerged cantilever beam to control fluid–structure interaction, utilizing harmonic motion and a semianalytical method to analyze its impact on thrust forces, lift forces, and hydrodynamic power dissipation, with implications for designing biomimetic underwater propulsion systems. Based on this work, (Devsoth and Pandey 2022) investigated the hydrodynamic forces in an array of cantilever beams oscillating in a viscous fluid, modeling fluid–structure interaction with symmetric and asymmetric shape morphing curvature and demonstrated the impact of frequency oscillation and shape parameters on damping, added mass effects, and hydrodynamic coupling, with implications for optimizing atomic force microscopy designs. Following their initial investigation, they extended their study to explore the impact of both single non-uniform microcantilever beams (Devsoth and Pandey 2023) and arrays consisting of both uniform and non-uniform microcantilever beams (Devsoth and Pandey 2024) on the computation of hydrodynamic forces. Recently, (Jujjuvarapu et al. 2024) fabricated hexagonal-shaped microcantilever beams and conducted both experimental and numerical frequency and damping analyses. They concluded that hexagonal beams provide better flexibility than their equivalent uniform beams, and drag force damping is the dominant energy dissipation mechanism. Razzaq et al. (2024) conducted a nonparametric method for identifying the nonlinear dynamics of a MEMS rectangular microcantilever beam subjected to electrostatic levitation forces, using the Restoring Force Surface method, Chebyshev polynomials, and experimental data, revealing nonlinearities such as velocity-dependent damping that parametric methods cannot capture, with results demonstrating robust predictive capability even under significantly increased voltage loads. Ilyas et al. (2016) investigated the static and dynamic behavior of electrostatically coupled laterally actuated silicon rectangular microbeams, revealing a significant drop in pull-in values compared to single microbeams, with potential applications in resonant sensing due to enhanced bandwidth and frequency shift near primary resonance.

Based on the above literature survey, it is evident that most research has predominantly concentrated on shape changes of conventional micro-cantilever beams through methods such as introducing perforations or adjusting beam width to enhance performance. However, limited research has been conducted on optimizing leaf-shaped micro-cantilever beams compared to the extensive work on optimizing conventional rectangular micro-cantilever beams. This indicates a gap in exploring alternative geometries for micro-cantilever beams and suggests an opportunity for further investigation into the potential benefits and applications of leaf-shaped designs in microsystems. Hence, in this work, we investigate the frequency and damping characteristics of MEMS cantilever-based microbeams with different shapes generated by optimizing the conventional rectangular uniform beam, examining their design principles, fabrication techniques, and dynamic characterization. Through a comprehensive analysis, we aim to provide insights into the influence of shape changes on the performance and functionality of these beams, paving the way for further advancements in MEMS cantilever beam technology.

The structure of the paper is outlined as follows: Sect. 2 explores the design process, detailing the generation of various beams, including diamond, hexagonal, and leaf-shaped microcantilever beams derived from the conventional uniform beam with specific dimensions. Section 3 focuses on the microfabrication process and experimental characterization of the proposed leaf-shaped beams. Moving forward, sect. 4 presents the comprehensive numerical modeling technique developed using FEM-based Coventorware numerical tools. Section 5 presents the results and discussions pertaining to frequency tuning and damping characteristics. Subsequently, we explore the potential application of these beams in MEMS accelerometers. Finally, sect. 6 concludes the paper, summarizing the findings and suggesting developments for further research.

## 2 Microbeam design

In this section, we discuss the designing of various micro-cantilever beams, achieved through the systematic exploration of sweeping angles $\theta$ and $\theta_1$. We study the important aspects of micro-cantilever beam design, exploring the effects of varying angles $\theta$ and $\theta_1$ on vibrational characteristics such as frequency, damping, and functional applications. Through theoretical analysis, computational modeling, and experimental validation, we aim to provide valuable insights into the design methodologies and optimization strategies for micro-cantilever beams for various applications. Consider a uniform-stepped micro-cantilever beam illustrated in Fig. 1a. This beam comprises step lengths $L_1$ and $L_2$, with a width of $w_s$ at the fixed end and $w_b$ at the free end. The total length of the beam is denoted as $L$. Accordingly, we systematically explored the design of micro-cantilever beams by varying the angles $\theta$ $(0^\circ \leq \theta \leq 180^\circ)$ and $\theta_1$ $(0^\circ \leq \theta_1 \leq 90^\circ)$ to design a wide range of micro-cantilever beams. These included uniform rectangular beams with a width of $w_s$, diamond-shaped, hexagonal-shaped, and uniform rectangular beams with a width of $w_b$, as shown in Fig. 1b. To provide a comprehensive view of our geometric exploration, we have denoted all the designed micro-cantilever beams using dotted lines with different colors, as depicted in Fig. 1b. Moreover, we developed a distinctive leaf-shaped microcantilever beam through careful polynomial approximations to the hexagonal beams, illustrated in Fig. 1c. This leaf-shaped beam has the same dimensions as the uniform-stepped micro-cantilever beam, ensuring consistency in structural parameters. This design iteration highlights our exploration of novel geometries derived from fundamental shapes, presenting opportunities for unique vibrational characteristics and potential applications in various engineering fields.

A detailed description of the microfabrication method used to fabricate the unique leaf-shaped micro-cantilever beams will be covered in the next section. A precise and straightforward procedure that allows for the accurate realization of these unique structures is silicon wet bulk micromachining, which is used in this method. To provide a detailed understanding of the process involved in fabricating these special micro-cantilever beams, we shall explain each stage of silicon wet bulk micromachining.

## 3 Microfabrication and experimental procedure

In this study, we used silicon wet bulk micromachining to create thermally grown silicon dioxide ($SiO_2$) microcantilever beams approximately 1 $\mu m$ thick. We began with a 4-inch diameter, p-type silicon wafer with a {100} orientation, polished on one side. A thermally grown oxide film acted as both a structural and masking layer during fabrication. The process involved depositing 1 $\mu m$ thick $SiO_2$ layer through thermal oxidation, followed by UV photolithography patterning with positive photoresist as a mask. Buffered hydrofluoric acid selectively etched the oxide layer. The wafer was diced into 2x2 $mm^2$ chips and cleaned with acetone and piranha solution, then rinsed in deionized water. A thin chemical oxide layer was grown to delay silicon etching, and the samples were immersed in 1% hydrofluoric acid to remove it. Microstructures were fabricated using wet bulk micromachining in a Tetramethylammonium hydroxide (TMAH) solution at $74.5 \pm 0.5^\circ$. After etching, the chips were rinsed and dried.

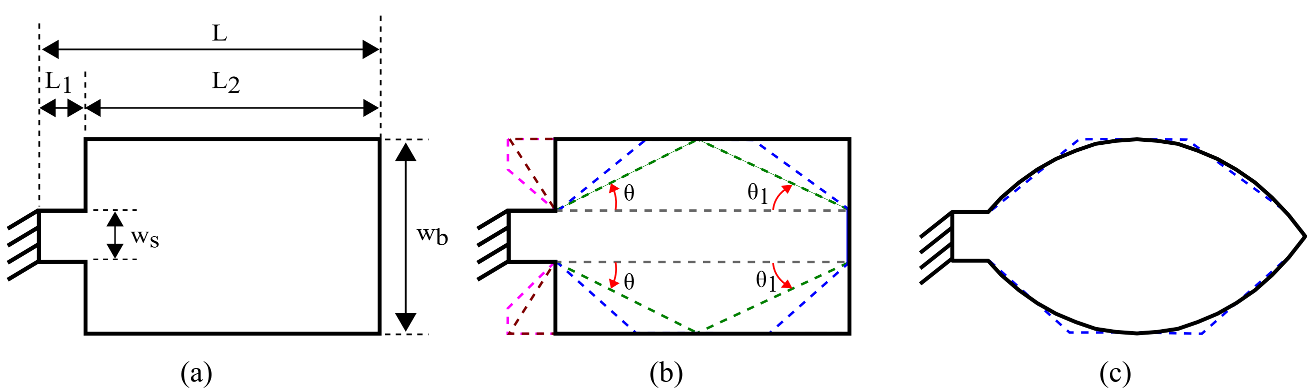



**Fig. 1** Schematics of (**a**) uniform-stepped micro-cantilever beam, (**b**) design of different micro-cantilever beams by sweeping the angles $\theta$ and $\theta_1$, and (**c**) Leaf based micro-cantilever beam

The thickness of the oxide layer was measured as 0.965 $\mu m$ using ellipsometry. For a more detailed description of the silicon wet bulk micromachining process, refer to the work of Jujjuvarapu et al. (2024)

The fabricated micro-cantilever beams are experimentally characterized using a Micro System Analyser (Polytec MSA-500) as shown in Fig. 2a. The laser-Doppler vibrometer precisely measures vibration velocity and displacement by detecting frequency shifts in backscattered light from a moving microstructure. This shift enables the measurement of velocity components along the axis of the laser beam. In the present study, our experimental procedure involves several steps. Initially, the chip containing the micro-cantilever beams is affixed onto a piezo actuator for excitation. The entire assembly is enclosed within an acrylic chamber to protect the setup from external influences, such as dust or humidity. This chamber is then positioned on the top of a platform capable of moving along the X, Y, and Z axes, facilitating precise positioning of the microbeams. A vibration isolation table is utilized to stabilize the experimental setup and further mitigate any external vibrations. The laser source emits a laser beam directed onto the micro-cantilever beams through an optical lens with a known frequency and phase shift. Subsequently, we define the scan points across the entirety of the beam surface and configure the acquisition settings in the software to meet the specific requirements for device characterization. Upon excitation by the function generator, the piezo actuator induces random vibrations in the microbeams. As the microbeams oscillate in an out-of-plane motion, they scatter the incident laser beam at each scan point, resulting in frequency and phase shifts in the reflected beam. A controller then compares the incident and reflected laser beams, generating output as a frequency response spectrum, as illustrated in Fig. 2b. This spectrum provides valuable insights into the dynamic behavior of the micro-cantilever beams under varying excitation conditions.

Further, we use the obtained frequency response spectrum to compute the frequency and damping characteristics of the micro-cantilever beams, such as natural frequency ($f_n$) and corresponding quality factor ($Q$) associated with a particular oscillation mode. Figure 2b, shows the frequency response spectrum and the half-power bandwidth method to measure the quality factor ($Q = f_n/(f_{\max} - f_{\min})$). The frequency corresponds to the peak amplitude ($A_{\max}$), giving the natural frequency ($f_n$) of the oscillating micro-cantilever beam.

In the following section, we discuss the numerical modeling of designed micro-cantilever beams in a finite element method-based numerical tool called CoventorWare®(Source yyy).

## 4 Numerical modeling

As previously discussed, determining the natural frequency and quality factor constitutes critical aspects in enhancing the performance of MEMS cantilever beams. In our study, we employed numerical modeling techniques within the finite element method framework using CoventorWare® software to assess experimental outcomes. The modeling process in Coventorware encompasses three primary stages: preprocessing, analysis, and post-processing. Initially, preprocessing involves material selection, where we designated thermally grown $SiO_2$ as the structural material and specified its corresponding material properties. Subsequently, we constructed a process architecture within the process editor to simulate the microfabrication of designed micro-cantilever beams with a defined structural thickness

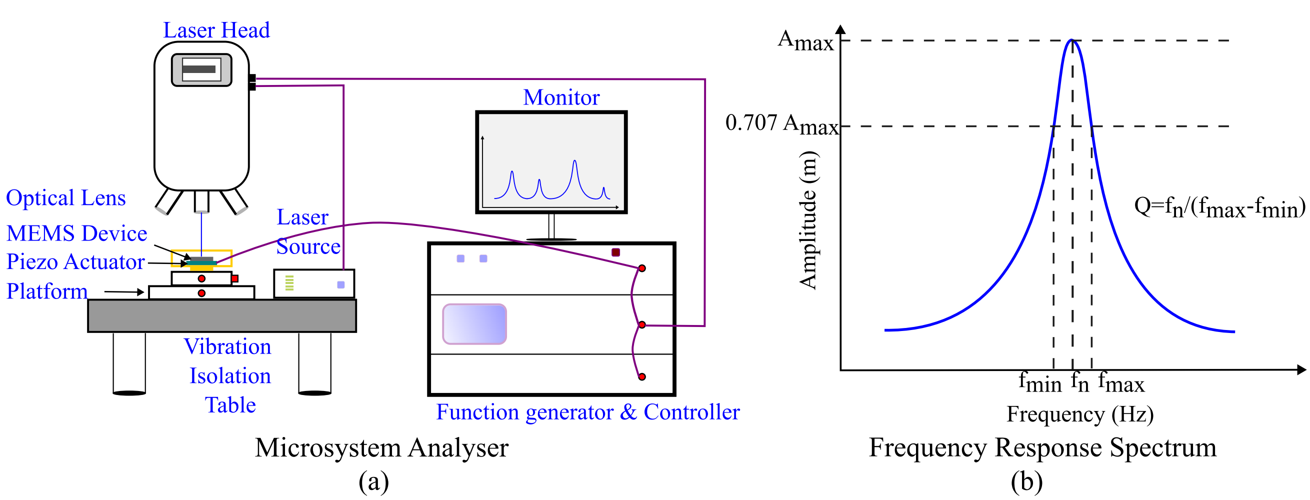



**Fig. 2** **a** Schematic of the experimental set up (Polytec, MSA-500), **b** half-power bandwidth method

of 0.965 $\mu m$. The subsequent step involves creating a 2D layout of the proposed designs in the layout editor by selecting appropriate design layers established in the process editor. Through this process, we generated a 3D model of the microcantilever beams with a thickness of 0.965 $\mu m$. Boundary conditions defining the cantilever, such as fixing one end while allowing oscillation at the other, were established on the created 3D model. Further, meshing was performed utilizing extruded bricks employing a hex-dominant algorithm, incorporating parabolic elements with 27 nodes, with each node containing 6 degrees of freedom.

During the analysis phase, we initially employed the MemMech solver incorporating the Lanczos algorithm. This solver facilitated the determination of modal frequencies, generalized mass, and mode shapes of the designed micro-cantilever beams. Additionally, to compute the quality factor associated with drag-force damping and squeeze-film damping effects, we utilized the DampingMM solver. This solver was employed with suitable displacement and velocity boundary conditions, enabling accurate estimation of the quality factor for the MEMS cantilever beams. The required numerical simulations have been carried out, and the results were analyzed. In the post-processing phase, we determined the natural frequencies and mode shapes corresponding to the first two transverse (out-of-plane) modes and the torsional mode of oscillation. Additionally, we assessed the quality factor associated with the first transverse mode of oscillation by computing the coefficients of drag $(C_d)$ and squeeze-film damping $(C_s)$. These coefficients enabled the quantification of drag-force damping and squeeze-film damping effects, respectively, contributing to the overall quality factor of the MEMS cantilever beam. When calculating the squeeze-film damping coefficient $(C_s)$, we used a trench depth of 80 $\mu m$, replicating the process of anisotropic trenching in silicon-wet bulk micromachining. This choice aligns with established methodologies for simulating realistic fabrication conditions and accurately assessing the squeeze-film damping effects in MEMS cantilever beams. Figure 3, llustrates the workflow within the CoventorWare software, which is based on the finite element method, used for comprehensive numerical modeling of the various beam configurations developed in this study. The detailed description and analysis of numerical modeling procedure has been presented in the Appendix I of supplementary material.

In the following section, we explain the results associated with modal analysis, static displacement, and damping of the designed microcantilever beams with various geometries.

## 5 Results and discussion

In this section, we discuss the numerical and experimental results of natural frequency, stiffness, and quality factors of various beams designed for the angles $\theta$ and $\theta_1$. All the numerical analyses have been carried out in CoventorWare® with the materials properties and dimensional parameters, which are given in Table 1. To enhance comprehension of our findings, we divided our discussion into two cases. In Case I, we explored the frequency tuning and associated drag and squeeze film damping of the first transverse mode across different beams. This analysis is achieved by varying the angles $\theta$ and $\theta_1$. Conversely, Case II examines leaf-shaped micro-cantilever beams, where we explore their unique characteristics and behavior regarding natural frequency and damping.

### 5.1 Modal analysis of different-shaped microcantilever beams

As previously mentioned, we initiated our investigation by analyzing a fundamental stepped uniform rectangular microcantilever beam with predetermined dimensions and angles, as depicted in Fig. 1a. Subsequently, we explored various beam configurations by systematically adjusting the angles $\theta$ and $\theta_1$, as demonstrated in Fig. 4a. Our exploration began with a straightforward rectangular uniform cantilever beam, with a width of 20 $\mu m$, where both $\theta$ and $\theta_1$ were set to 0 $(\theta = \theta_1 = 0)$. Following this, we investigated diamond-shaped beams within the range of $0 < \theta = \theta_1 \leq 18.43°$, categorized as category I. Following the design of category-I beams, we defined hexagonal-shaped beams characterized by angles within the range $18.43° < \theta = \theta_1 < 90°$, designated as category II. It is noteworthy that when $\theta = \theta_1 = 90°$, the beam reverts to the fundamental stepped uniform rectangular microcantilever configuration. In this instance, the width at the fixed end $(w_s)$ remains 20 $\mu m$, while the width at the free end $(w_b)$ expands to 80 $\mu m$. Moreover, through systematic angle adjustments, ranging from $90° < \theta < 123.69°$ while maintaining $\theta_1 = 90°$, we could generate beams classified under category III. Following this, we generated beams categorized as category IV by maintaining the angles within the range of $123.69° < \theta < 180°$ and $\theta_1 = 90°$. Notably, in the case of category IV, we applied two distinct boundary conditions, denoted as $BC_1$ and $BC_2$. Under $BC_1$, all surfaces to the left of the beam were fixed, while under $BC_2$, only the central step beam was fixed. This approach allowed for a comprehensive examination of beam behavior under varied boundary conditions. Ultimately, by fixing angles, specifically $\theta = 180°$ and $\theta_1 = 90°$, we successfully attained a uniform rectangular cantilever beam

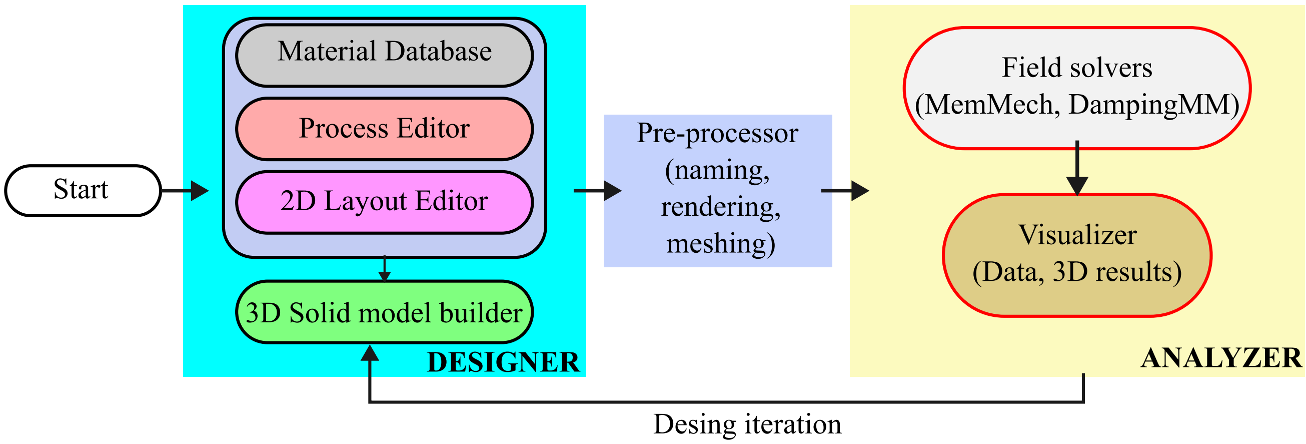



**Fig. 3** Work flow in the CoventorWare for the numerical modeling of leaf-shaped microcantilever beams

**Table 1** Material properties and dimensional parameters of $SiO_2$ microcantilever beams

| S.No | Parameter | Value |
|---|---|---|
| 1 | Young's Modulus (E) | 66.26 GPa |
| 2 | Poisson's ratio ($v$) | 0.28 |
| 3 | Density ($\rho$) | 2200 $kg/m^3$ |
| 4 | Structural thickness (t) | 0.965 $\mu m$ |
| 5 | Step length ($L_1$) | 20 $\mu m$ |
| 6 | Step length ($L_2$) | 180 $\mu m$ |
| 7 | Total length of the beam ($L$) | 200 $\mu m$ |
| 8 | Width at fixed end of the beam ($w_s$) | 20 $\mu m$ |
| 9 | Width at free end of the beam ($w_b$) | 80 $\mu m$ |

with a width of 80 $\mu m$. This configuration represents a fundamental and straightforward structural arrangement for further investigation and comparison within our study. For better visualization, all the beam configurations, along with the angle constraints, are shown in Fig. 4a.

The numerical modeling of our proposed designs was conducted utilizing Coventorware® software. In Fig. 4b, we present the first transverse mode natural frequency ($f_1$) of various beams plotted against the angle ($\theta$). To facilitate analysis, we divided the frequency response of the beams into four distinct zones, denoted as I, II, III, and IV. This division aids in understanding the behavior of the beams across different angular ranges. It is noted that the value of $f_1$ corresponds to the case of a uniform rectangular beam at $\theta = 0$, which is 21.512 kHz. Whereas in zone I,($0 < \theta < 18.43^\circ$), which corresponds to the diamond shape beams, the value of $f_1$ decreases. This decrease is attributed to an increase in the effective mass of the beams, resulting from the transition in geometry from a uniform beam to a diamond one. Likewise, in zone II ($18.43^\circ < \theta < 90^\circ$), representing hexagonal-shaped beams, there is an exponential decrease in the value of $f_1$. This decline is due to the increased effective mass of the hexagonal beams. Furthermore, the frequency value of the stepped uniform beam ($\theta = 90$) is determined to be 14.221 kHz, highlighting its distinct vibrational characteristics within this range of angles. However, In Zone III ($90^\circ < \theta < 123.69^\circ$), we observe a relatively constant value of $f_1$. This is because of constant effective stiffness and effective mass values for the chosen beam dimensions. This finding suggests that within this angular range, the vibrational characteristics of the beams remain relatively unchanged, offering insights into their behavior under these specific conditions. Furthermore, In zone IV ($123.69^\circ < \theta < 180^\circ$), we can observe the influence of two kinds of boundary conditions $BC_1$, $BC_2$ over the value of $f_1$ corresponds to the beams in category IV. By implementing boundary condition $BC_1$, the effective stiffness of the beams increases while the effective mass decreases. As a result, the beams exhibit a higher natural frequency until they reach 21.795 kHz, corresponding to the uniform cantilever beam ($\theta = 180^\circ$) with a width of 80 $\mu m$. Additionally, we explored the impact of $BC_2$ on the first transverse mode frequency ($f_1$), revealing that the natural frequency of the beams becomes constant. This is due to the consistent values of effective stiffness and mass across the beam configurations within the specified angle range. These findings highlight the significance of boundary conditions in determining the vibrational characteristics of the beams, emphasizing the need for careful consideration when designing microcantilever structures for specific applications.

Additionally, we conducted analyses to determine the frequencies of the second transverse mode of oscillation ($f_2$) and the torsional mode of oscillation ($f_3$), as illustrated in Fig. 4c, d, respectively. From the results presented in Fig. 4c, we observed that $f_2$ increases as the angle $\theta$ increases up to 10°. This increased trend can be attributed

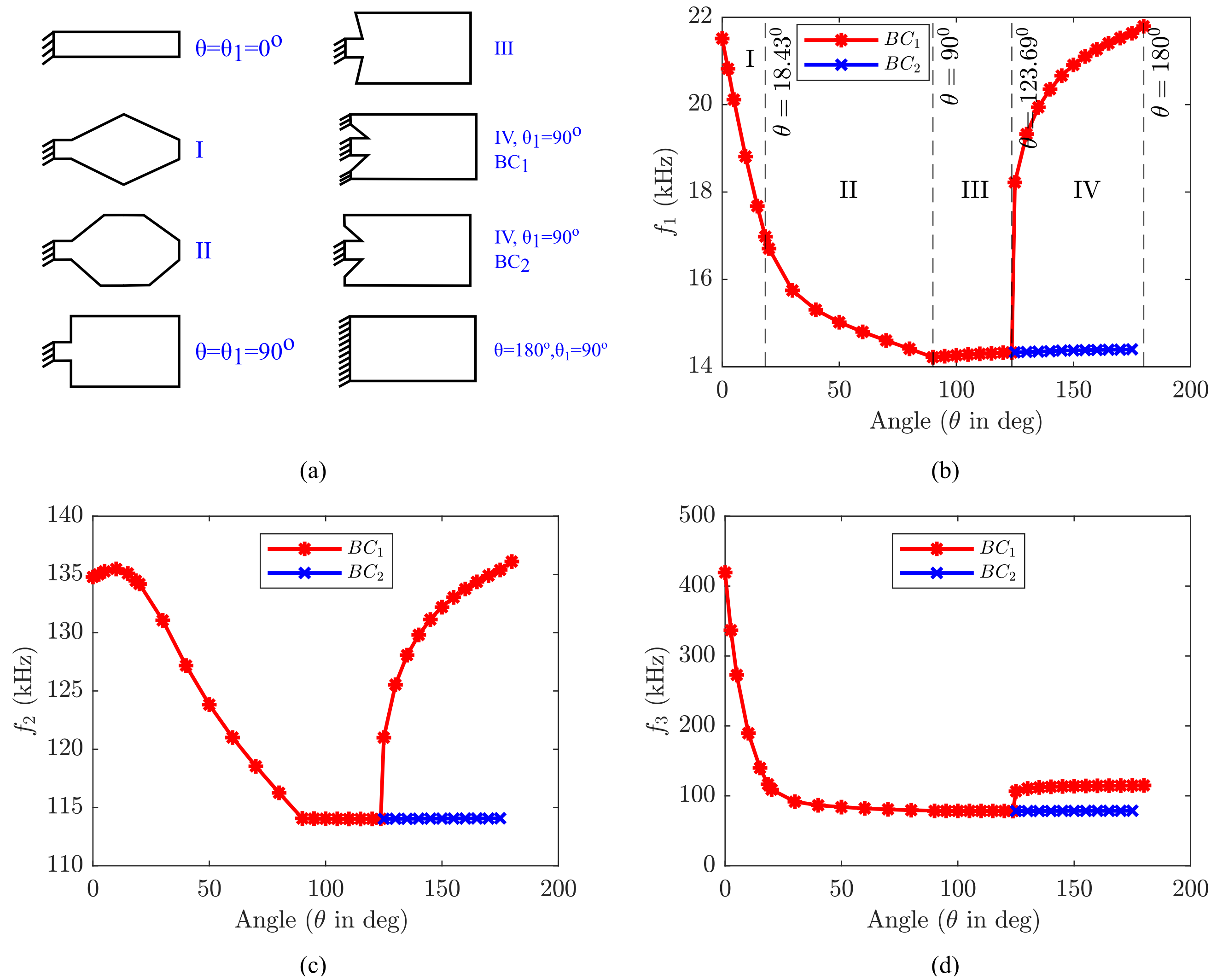



**Fig. 4** **a** schematic of various beams generated through angle ($\theta$) sweeping from 0° to 180°, **b** first transverse mode natural frequency vs angle, **c** second transverse mode natural frequency vs angle, and **d** torsional mode natural frequency vs angle

to the increase in the effective stiffness of the beams, particularly affecting the second transverse mode of oscillation. Subsequently, as shown in the same figure, there is a subsequent decrease in the value of $f_2$ as $\theta$ approaches 90°. This decreasing trend results from the increased effective mass of the hexagonally-shaped beams, further influencing their vibrational behavior. Further, for the angle $90^\circ < \theta < 123.69^\circ$, $f_2$ is constant, which suggests that the beams of category III have constant values of effective stiffness and mass. Similarly to $f_1$, the frequency $f_2$ increases for category IV beams subjected to $BC_1$. This increase in $f_2$ can be attributed to the concurrent increase in effective stiffness and decrease in the effective mass of the beams. This change in the structural properties leads to a higher natural frequency for the second transverse mode of oscillation ($f_2$), emphasizing the influence of boundary conditions on the dynamic behavior of micro-cantilever beams. Whereas the beams with $BC_1$ have a constant value of $\approx 114 kHz$. This suggests that Category III and IV beams subjected to $BC_2$ exhibit minimal impact on the natural frequency, whereas those subjected to $BC_1$ undergo significant frequency tuning for the specified dimensions of the microcantilever beams. Subsequently, the phenomenon can be seen in the torsional mode of oscillations as shown in Fig. 4d.

## 5.2 Static displacement analysis of different-shaped microcantilever beams

Following the modal analysis, we conducted a static displacement analysis of the proposed designs under normal loads ranging from 1 nN to 10 nN, applied at the free end of the beams. Figure 5, illustrates the displacement characteristics of the beams for different angles ($\theta$), considering

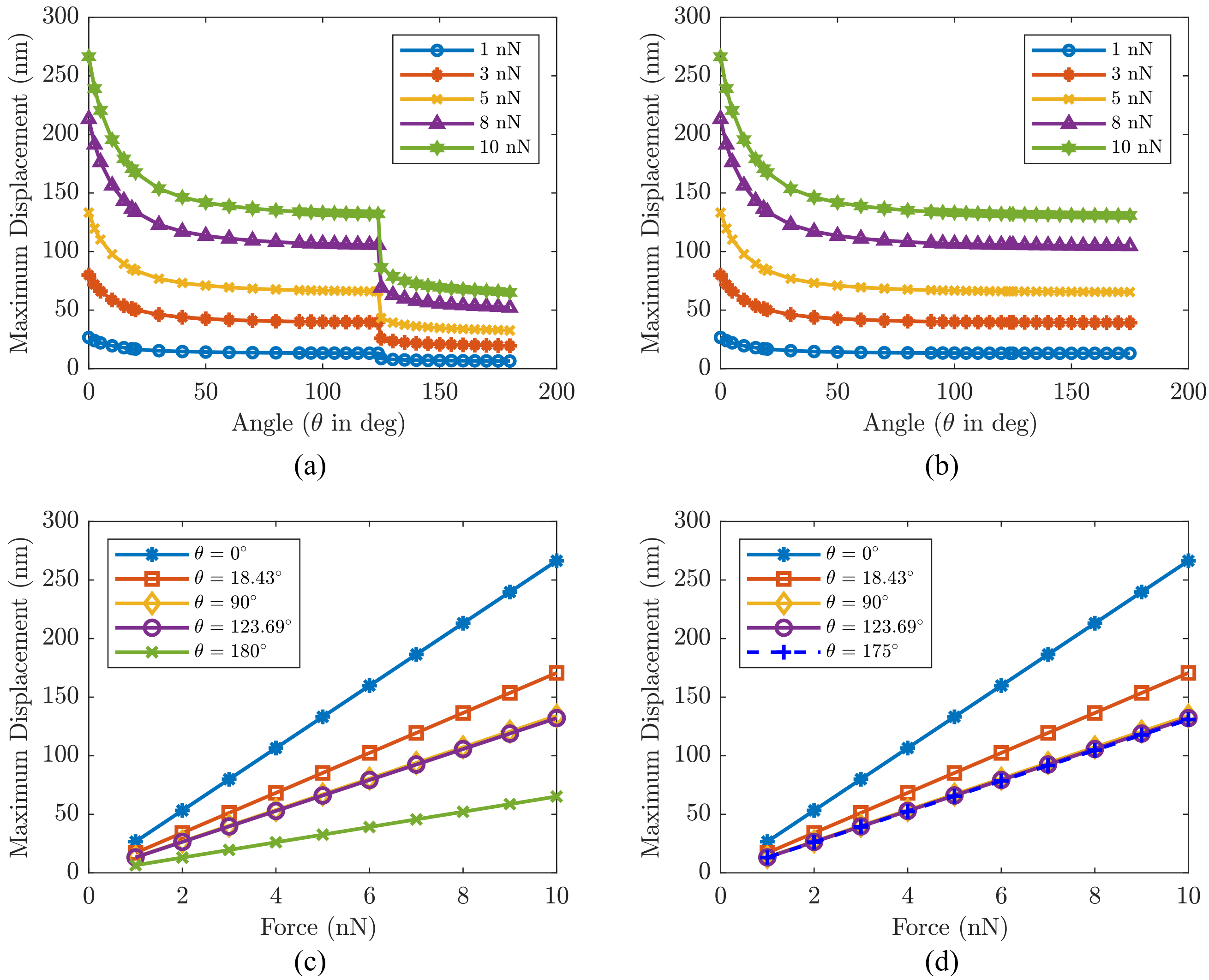



**Fig. 5** (**a**) and (**b**) depict the maximum displacement (in nm) vs the angle ($\theta$) for boundary conditions $BC_1$ and $BC_2$, respectively. (**c**) and (**d**) illustrate the maximum displacement (in nm) vs the force (in nN) for various beam configurations represented by $\theta$, respectively

two boundary conditions: $BC_1$ and $BC_2$. This analysis provides insights into how the beams respond to varying loads and boundary conditions, offering valuable information for understanding their mechanical behavior and structural integrity. In Fig. 5a, we examine the maximum displacement of various beams subjected to normal loads ranging from 1 nN to 10 nN. Initially, as the angle $\theta$ increases within the range of $0 \leq \theta \leq 123.69°$, a consistent trend is observed: the maximum displacement steadily decreases. However, beyond $\theta = 123.69°$ up to $\theta = 180°$, a notable decrease in the maximum displacement occurs. This sudden decrease is attributed to the stiffening effect induced by implementing $BC_1$. Essentially, the utilization of $BC_1$ leads to a significant enhancement in the stiffness of the beams within this angular range, thereby influencing their mechanical response. It has been noted that there is a consistent reduction in displacement for beams employing $BC_2$, as depicted in Fig. 5b. This consistent decrease in displacement suggests a stable mechanical response under varying boundary conditions. The implementation of $BC_2$ appears to contribute to the overall stiffness of the beams, resulting in a more controlled and predictable displacement behavior.

Additionally, in Fig. 5c, we present the maximum displacement encountered by beams of different shapes, represented by the angle ($\theta$), under normal loads ranging from 1 to 10 nN, while subjected to $BC_1$. The results indicate that all proposed beams exhibit linear displacement characteristics across the range of loading. Notably, among the various shapes analyzed, a uniform beam ($\theta = 0°$) with a width of 20 $\mu m$ demonstrates a 56.073% higher displacement than a diamond-shaped beam ($\theta = 18.43°$). This can be attributed to the lower transverse stiffness of the uniform beam, which allows for greater deflection under the applied load. Furthermore, it is observed that there is a distinct order of decreasing displacement under the given load conditions: first, the uniform beam with a width of 20 $\mu m$ and $\theta = 0°$ exhibits the highest displacement, followed by the diamond-shaped beam with $\theta = 18.43°$, then the stepped uniform beam with $\theta = 90°$, and finally, the uniform beam with a width of 80 $\mu m$ and $\theta = 180°$ shows the lowest displacement. This trend can be attributed to the increase in stiffness of the beams when subjected to the boundary condition $BC_1$. Essentially, the higher stiffness restricts the deformation of the beams, resulting in reduced displacement. A similar trend can seen for the beams with $BC_1$, which is illustrated in the Fig. 5(d). Hence, these findings provide the importance of considering beam geometry and stiffness properties in optimizing the displacement characteristics of microcantilever structures for specific applications.

Based on the previous discussion about frequency tuning and displacement analysis, it is essential to determine the effective stiffness and mass for the different beams under the given boundary conditions. From Hooke's law, the effective stiffness of a beam can be defined as (1):

$$F = K_{\text{eff}} \times \delta \Rightarrow K_{\text{eff}} = \frac{F}{\delta} \tag{1}$$

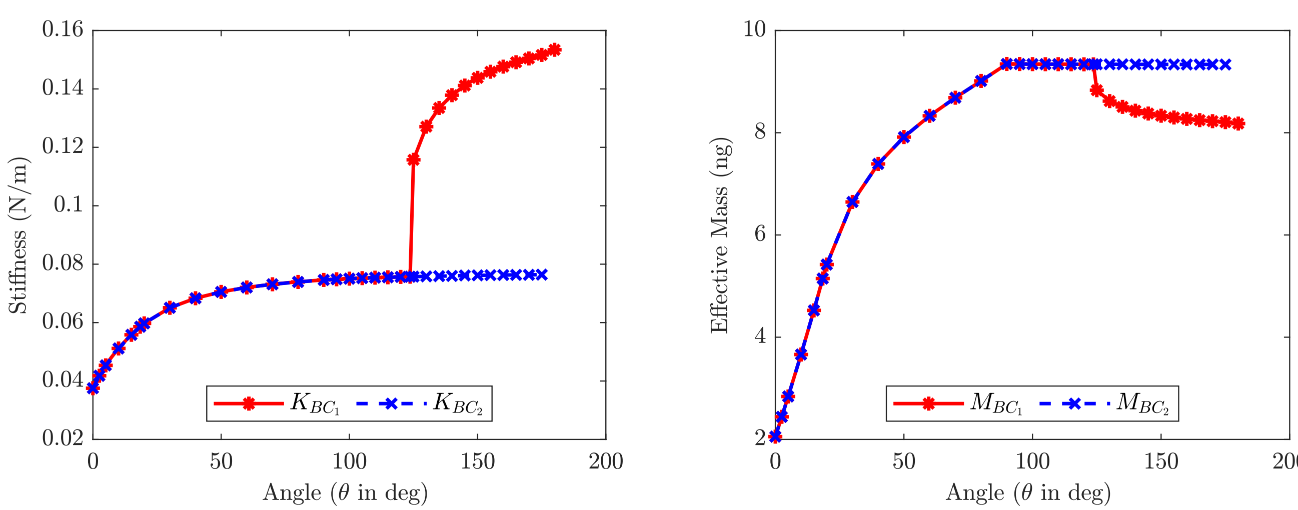
here, $F$ is the applied normal load and $\delta$ is maximum displacement of the beam.

$$f_1 = \frac{1}{2\pi}\sqrt{\frac{K_{\text{eff}}}{M_{\text{eff}}}} \Rightarrow M_{\text{eff}} = \frac{(2\pi f_1)^2}{K_{\text{eff}}} \tag{2}$$

After substituting the values of $F$ and $\delta$ obtained from the previous static displacement analysis into Eq. (1), we calculated the effective stiffness ($K_{\text{eff}}$) for various beam configurations across the sweeping angle ($\theta$), as depicted in Fig. 6a. Our findings reveal that the value of $K_{\text{eff}}$ increases until the angle $\theta$ reaches 123.69° for both the boundary conditions. Furthermore, beams subjected to $BC_1$ exhibit an increasing stiffness with the angle, while those under $BC_2$ maintain a consistent stiffness across the angle range of $123.69° < \theta < 180°$. This observation suggests that the choice of boundary condition significantly influences the mechanical behavior of the beams. Under $BC_1$, the increasing stiffness can be attributed to the constraints imposed on the beam structure, leading to greater resistance to deformation as the angle increases. Conversely, beams under $BC_2$ experience uniform conditions, resulting in a constant stiffness regardless of the angle. This distinction gives the importance of selecting appropriate boundary conditions to achieve desired mechanical properties in microcantilever beam designs. We proceeded to compute the effective mass ($M_{\text{eff}}$) for beams of various shapes, as illustrated in Fig. 6b. Similar to the trend observed for $K_{\text{eff}}$, the value of ($M_{\text{eff}}$) increases until the angle ($\theta$) reaches 123.69°. However, in the angle range $123.69° < \theta < 180°$, the behavior of ($M_{\text{eff}}$) differs from that of $K_{\text{eff}}$. Instead of continuing to increase for the $BC_1$, ($M_{\text{eff}}$) exhibits an opposite trend. Whereas the beams with $BC_2$ have a constant value for the given angle. This change in behavior indicates a shift in the structural characteristics of the beams as the angle increases.

## 5.3 Validation of numerical approach

Additionally, we verified the accuracy of our numerical approach by comparing various parameters with the analytical modeling of a uniform cantilever beam. Table 2




(a) (b)

**Fig. 6** Numerically computed (**a**) The effective stiffness and (**b**) the effective mass of different beams with respect to the sweeping angle ($\theta$) for two boundary conditions $BC_1$ and $BC_2$

**Table 2** Comparision of analytical and numerical values of displacement ($\delta$), effective stiffness ($K_{\text{eff}}$), effective mass ($K_{\text{eff}}$), and first transverse mode natural frequency of uniform cantilever beam ($\theta = 0^{\circ}$) with $L = 200\mu m$, $w_b = 20\mu m$, and a thickness ($t$) of 0.965 $\mu m$. Note, here $M_b = \rho \times L w_b t$, defines the total mass of the microcantilever beam

| | $\delta = \frac{3EI}{L^3}$ (nm) | $K_{\text{eff}}$ (N/m) | $M_{\text{eff}} = 0.26 M_b$ (ng) | $f_1$ (kHz) |
|---|---|---|---|---|
| Ana | 26.896 | 0.0372 | 2.207 | 21.378 |
| Num | 26.637 | 0.0375 | 2.116 | 21.512 |
| Error (%) | 0.96 | 0.81 | 4.12 | 0.62 |

presents a comparison of analytical and numerical values for displacement ($\delta$), effective stiffness ($K_{\text{eff}}$), effective mass ($M_{\text{eff}}$), and the first transverse mode natural frequency of a uniform cantilever beam ($\theta = 0^{\circ}$) with dimensions $L = 200\mu m$, $w_b = 20\mu m$, and a thickness ($t$) of 0.965 $\mu m$. The obtained results have shown a greater agreement with each other for the chosen dimensions of the uniform cantilever beam. This comparison enables us to assess the accuracy and reliability of our numerical simulations in predicting the behavior of microcantilever beams under various loading and boundary conditions.

### 5.4 Damping analysis of different shaped microcantilever beams

Drag and squeeze-film damping are two common forms of damping observed in microcantilever beams. Both drag and squeeze-film damping play crucial roles in determining the dynamic behavior and performance of microcantilever beams, especially in applications such as sensors, actuators, and resonators. Understanding and quantifying these damping mechanisms are essential for optimizing the design and functionality of MEMS devices. In our study, we utilized numerical methods to calculate the coefficients of drag force damping ($C_d$) and squeeze film damping ($C_s$). We employed the DampinMM solver within the Coventorware® software. This solver facilitates the computation of these damping coefficients, which are essential parameters in analyzing the damping characteristics of microcantilever beams. The quality factor associated with these damping mechanisms can be calculated from the following expression:

$$Q = \frac{M_{\text{eff}} * \omega_1}{C} \tag{3}$$

Here, we denote the coefficient of drag force damping as $C = C_d$, with the corresponding quality factor represented as $Q = Q_{\text{drag}}$. Similarly, for squeeze-film damping, we use $C = C_s$, and the associated quality factor is denoted as $Q = Q_{\text{squeeze}}$. Additionally, $M_{\text{eff}}$ is the effective mass, $\omega_1 = 2\pi f_1$ defines the angular frequency of the first transverse mode of oscillation.

The numerical results depicting drag and squeeze-film damping, along with their respective quality factors, are presented in Fig. 7. In Fig. 7a, the coefficient of drag force damping ($C_d$) is illustrated for various beams under two distinct boundary conditions, as well as for an equivalent uniform beam (EUB) compared to diamond-shaped microcantilever beams. The obtained data reveals that $C_d$ increases as the angle ($\theta$) approaches 90° for both boundary conditions, $BC_1$ and $BC_2$, owing to the expanding surface area of the beams. Subsequently, $C_d$ is constant for $BC_1$ as the beams maintain a nearly constant surface area. Conversely, for $BC_2$, $C_d$ decreases within the angle range $123.69^{\circ} < \theta < 180^{\circ}$. This reduction is attributed to the increased frequency caused by the stiffing of the beams, which limits the interaction time between air particles and the structure surface, reducing the damping effect. Further, it is noteworthy that the value of $C_d$ for equivalent uniform beams to the diamond-shaped beams ($0^{\circ} < \theta \leq 18.43^{\circ}$) exhibits an increasing trend, which can be attributed to the increasing surface area. However, this becomes constant within the angle range ($18.43^{\circ} \leq \theta \leq 180^{\circ}$), as the surface area of the beams remains constant in this specified angle range. The quality factor associated with the drag force damping has been shown in Fig. 7b. From this plot, it has been observed that the $Q_{\text{drag}}$ of diamond-shaped beams increases as the angle increases from $0^{\circ} < \theta \leq 18.43^{\circ}$. This is because the numerator value in the equation (3) is dominant over the denominator value, exerting dominance over the overall value of $Q_{\text{drag}}$. However, for the hexagonal-shaped beams ($18.43^{\circ} < \theta < 90^{\circ}$), the value of $Q_{\text{drag}}$ first increases and then decreases, due to the increasing surface causing the influence of $C_d$ is more dominant over the numerator in equation (3). Furthermore, the value of $Q_{\text{drag}}$ remains constant within the range of $90^{\circ} \leq \theta \leq 90^{\circ}$ because of the constant surface area of the beams within this particular angular range. In the case of $BC_2$, within the same angular range, the value of $Q_{\text{drag}}$ increases and eventually reaches a value of the uniform beam with a width of 80 $\mu m$. This increase can be attributed to the simultaneous increase in the numerator and decrease in the denominator of the Eq. (3). This suggests that beams subjected to $BC_2$ offer a superior quality factor compared to those under $BC_1$

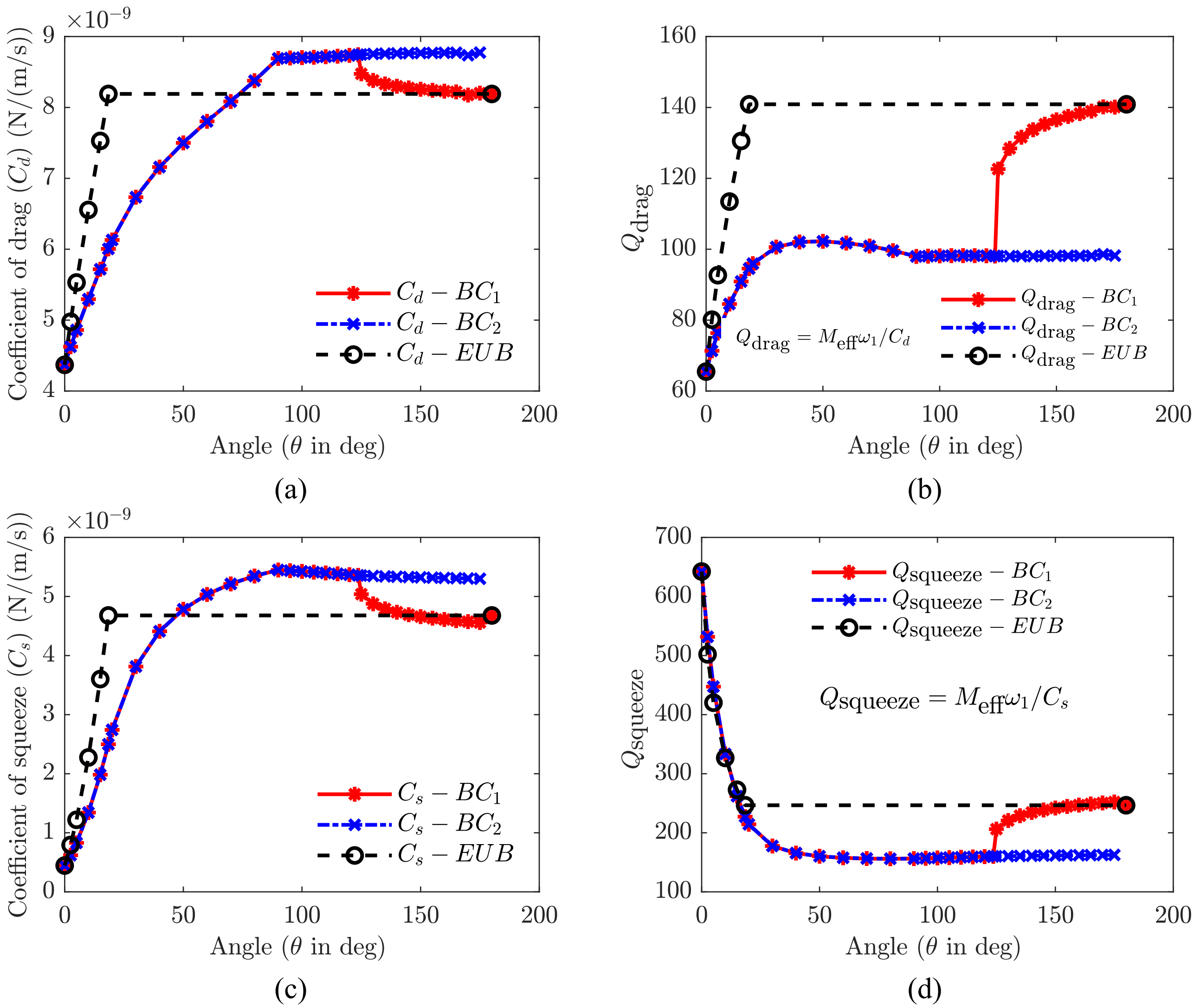



**Fig. 7** **a** Coefficient of drag ($C_d$), **b** quality factor corresponds to the drag force damping ($Q_{\text{drag}}$), **c** coefficient of squeeze-film damping ($C_s$), and **d** quality factor associated with squeeze film damping of first transverse mode of oscillation for various shaped beams which are represented by the angel($\theta$)

despite having identical geometrical dimensions. Subsequently, we calculated the $Q_{\text{drag}}$ for equivalent uniform beams resembling the diamond shape. It increases until the angle reaches 18.43°, after which it has a constant value of 140.888. This observation suggests that the equivalent uniform beams provide a higher quality factor than beams of varying shapes under the same dimensions.

Continuing our analysis, we determined the coefficient of squeeze-film damping ($C_s$) and its corresponding quality factor ($Q_{\text{squeeze}}$) while keeping the trench depth constant at 80 $\mu m$, as shown in Figs. 7c, d respectively. Figure 7c illustrates that $C_s$ follows similarly to $C_d$, exhibiting a decrease in magnitude. However, an opposite trend is observed in the value of $Q_{\text{squeeze}}$, as indicated in Fig. 7d. This deviation can be attributed to the increase in surface area of the beams as the angle ($\theta$) increases. The increased surface area leads to greater energy dissipation through squeeze film damping, resulting in a reduction in the associated quality factor. Notably, beams with $BC_2$ exhibit higher ($Q_{\text{squeeze}}$) values compared to those with $BC_1$ within the range of $123.69° \leq \theta \leq 180°$. This discrepancy arises from the dominance of the numerator over the denominator in equation (3). For comparison purposes, we also computed the ($Q_{\text{squeeze}}$) values for equivalent uniform beams (EUB) of diamond shape. Interestingly, the EUB demonstrates a reduction in the magnitude of the quality factor due to increased surface area.

In the subsequent section, we explain the parametric study concerning the modal, static displacement, and damping properties of leaf-shaped micro-cantilever beams. This study aims to comprehensively analyze and understand the various factors influencing these characteristics, providing important information related to the dynamics of these specialized structures.

### 5.5 Leaf-shaped microcantilever beams

Initially, we conducted the numerical modal analysis using Coventorware software for the beams depicted in Fig. 8a. Our first focus was on a solid leaf-shaped microcantilever beam, derived as an approximation from the hexagonally shaped beams discussed in sect. 2. This approach allowed us to explore the structural dynamics and resonance characteristics of the microcantilever beam design under investigation. In this present study, our primary emphasis was on investigating three variations of beams, denoted as Beam 1 and Beam 2, derived from a solid leaf beam design, and Beam 3, derived from an arrow beam design, respectively. When considering the solid leaf beam, we introduced a parameter denoted as $h$, which characterizes the lateral etching of the solid leaf beam relative to the edge of the stepped beam. Conversely, in the context of the arrow beam, the parameter $h$ describes the height of the webs measured from the edge of the arrow beam. This distinction in parameter definition reflects the unique geometric features and design considerations inherent in each beam type, offering insights into their respective fabrication processes and structural configurations.

Figure 8, illustrates the relationship between the first transverse mode natural frequency of the leaf-shaped beam and the parameter $h$. This graphical representation allows for a clear understanding of how changes in the parameter $h$ influence the vibrational characteristics of different microbeams. For Beam 1, as the parameter $h$ increases from 0 to 30 $\mu m$, it gradually approaches the configuration of Beam 3, resulting in a convergence of their structural characteristics. Concurrently, the corresponding first mode natural frequency experiences a steady increase, eventually

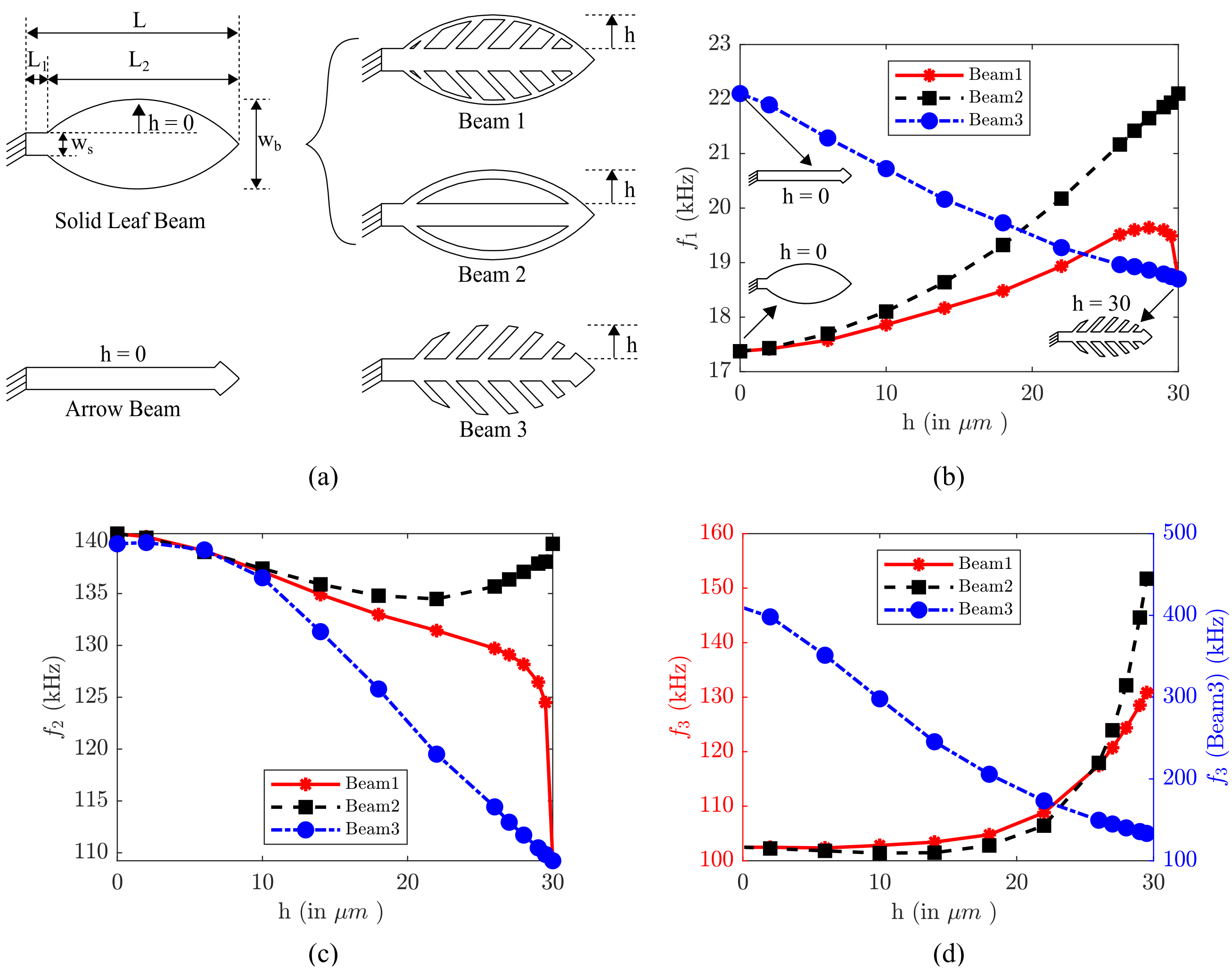



**Fig. 8** **a** Schematics of different leaf-shaped microcantilever beams, **b** first transverse mode, **c** second transverse mode, and **d** torsional mode frequency of different beam configurations

reaching the value observed for Beam 3. The observed trend in $f_1$ comes from the decrease in effective mass, which arises from etching away structural material. This reduction in effective mass leads to a corresponding increase in the natural frequency of the system. Likewise, in the case of Beam2, as the $(h)$ increases, the natural frequency $(f_1)$ increases until it reaches the value of the arrow beam. This increase in frequency can be attributed to the decreased effective mass for the Beam 2 configuration. Moreover, in the case of the Beam3 configuration, there is a decrease in the natural frequency $(f_1)$ as the web height $(h)$ increases. This decline can be attributed to increased effective mass from adding webs to the basic arrow beam shape. Additionally, we have presented the modes corresponding to the second transverse mode and torsional mode of oscillation for the designed beams in Fig. 8c, d, respectively. These modes offer insights into the dynamic behavior and structural response of the beams under various geometric conditions, aiding in the comprehensive analysis of their dynamic characteristics. Therefore, by tuning the geometric parameter associated with etching, denoted as $h$, one can finely tune the frequency of the proposed leaf-shaped designs to suit the specific requirements of the intended application. This capability offers a practical means of tailoring the dynamic characteristics of the beams to meet the desired performance criteria, enhancing their versatility and applicability across a range of sensing applications.

Proceeding further, we computed the numerical $Q_{\text{drag}}$ and $Q_{\text{squeeze}}$ of the leaf-shaped microcantilever beams, as shown in Fig. 9. The coefficient of drag $(C_d)$ has been shown in Fig. 9a, and the corresponding quality factor $(Q_{\text{drag}})$ is shown in Fig. 9b. The analysis reveals that the value of $Q_{\text{drag}}$ decreases for both Beam 1 and Beam 2 as the parameter 'h' increases from 0 to 30 $\mu m$. Conversely, for Beam 3, $Q_{\text{drag}}$ remains nearly constant across the same range of h. This suggests that the drag quality factor $Q_{\text{drag}}$ is significantly influenced by the presence of an outer periphery in the leaf-shaped beams (Beam1 and Beam2). In contrast, the provision of webs alone (Beam3) does not notably affect $Q_{\text{drag}}$.Therefore, the design and structure of the beams, particularly the outer periphery, are critical in determining the drag-induced damping characteristics. Future design optimizations for minimizing $Q_{\text{drag}}$ should focus on the outer periphery of the beams rather than internal webs alone.

Similarly, the coefficient of squeeze $(C_s)$ is depicted in Fig. 9c, with the corresponding quality factor $Q_{\text{squeeze}}$ illustrated in Fig. 9d. As observed with $Q_{\text{drag}}$, the value of $Q_{\text{squeeze}}$ increases for both Beam 1 and Beam 2, while remaining nearly constant for Beam 3.Therefore, the presence of an outer peripheral structure is a key factor in determining the damping behavior of leaf-shaped cantilever beams, affecting both drag-induced and squeeze film-damping mechanisms.

## 5.6 Experimental characterization of leaf-shaped microcantilever beams

Continuing our investigation, we conducted experimental characterization of the proposed leaf-shaped beams utilizing a Polytec-MSA500 laser doppler vibrometer, as illustrated in the schematic of Fig. 2. By following the standardized experimental procedure outlined in Sect. 2, we measured the first two transverse and torsional modes of oscillations, along with their respective quality factors, employing the half-power bandwidth method.

Figure 10a, shows the optical images depicting the fabricated leaf-shaped micro-cantilever beams, denoted as Leaf 1 through Leaf 5 configurations. Leaf 1 represents the solid leaf beam without any perforations, whereas Leaf 2 features a perforated beam with inclined webs containing ridges designed to bridge the gap between successive webs. Moreover, Leaf 3 is configured as a perforated beam featuring inclined webs with an outer periphery. At the same time, Leaf 4 incorporates an arrow beam design characterized by inclined webs with ridges and lacks an outer periphery. Lastly, Leaf 5 represents an arrow beam design with inclined ridges without an outer periphery. The beams have been arranged in increasing order based on their mass distribution relative to the given geometric changes and dimensions. These distinct configurations enable a detailed examination of the structural changes and performance attributes of the micro-cantilever beams, contributing to a comprehensive understanding of their dynamic behavior.

The experimental and numerical natural frequencies $(f_1)$ for the transverse mode of oscillation are presented in Table 3, while the corresponding numerical mode shapes are illustrated in Fig. 9b. Analysis of the results reveals a close agreement between the experimental and numerical frequency values, with a maximum percentage error of 11.167% observed for the Leaf 4 configuration. This consistency between the experimental and numerical findings validates the accuracy and reliability of the numerical simulations, affirming their utility in predicting the dynamic behavior of the leaf-shaped micro-cantilever beams. The difference between numerical and experimental values can arise due to factors such as material property variations, fabrication inconsistencies, boundary condition assumptions, and modeling limitations in capturing complex damping effects like squeeze-film damping. Environmental influences, such as temperature and humidity, and limitations in measurement instruments further impact these results. This work effectively predicts beam behavior within acceptable error margins, supported by statistical

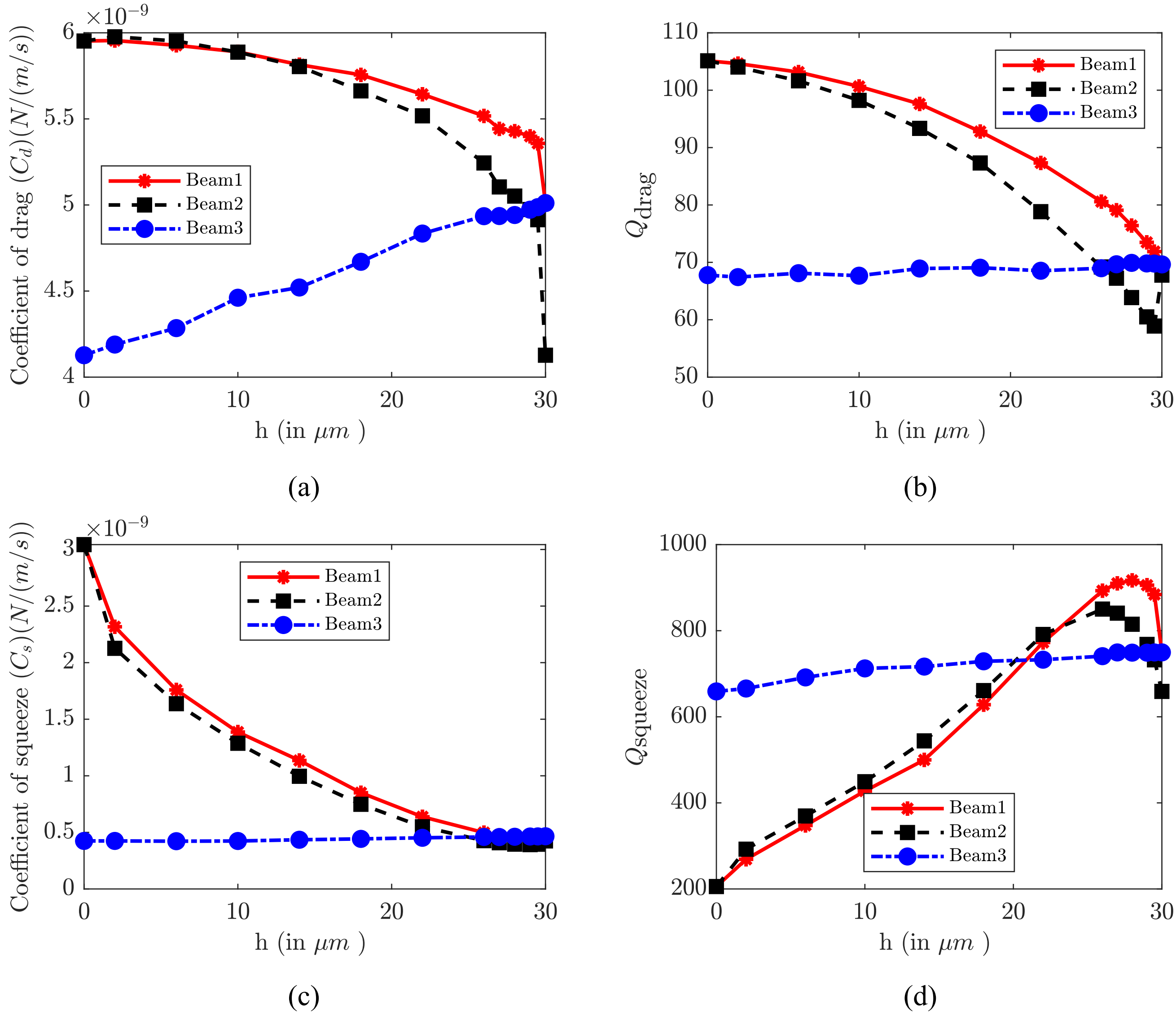



**Fig. 9** **a** Coefficient of drag ($C_d$), **b** drag force damping quality factor ($Q_{\mathrm{drag}}$), **c** Coefficient of squeeze-film ($C_s$), and **d** drag force damping quality factor ($Q_{\mathrm{squeeze}}$)

tools like the half-power bandwidth method for accurate quality factor estimation. These findings provides the accuracy of the chosen modeling approach for dynamic characterization of leaf-shaped microcantilever beams.

Following this, we measured the second transverse and torsional modes of the proposed beams. The obtained values are tabulated in Table 4, and their respective numerical mode shapes are depicted in Fig. 10a, b, respectively. It is observed that both experimental and numerical second transverse modes have shown a good agreement with each other at a maximum percentage error of 6.488%. Hence, This systematic tuning of natural frequency facilitates a clear understanding of how changes in geometry and dimensions impact the resonance and mass distribution of the beams, providing valuable insights into their structural characteristics and dynamic behavior Fig. 11.

Given the important role of quality factor analysis in comprehending damping behavior and optimizing the performance of MEMS cantilever beams, we conducted both experimental and numerical assessments. Experimentally, we determined the quality factor under ambient conditions using the half-power bandwidth method. Additionally, numerical simulations were employed to compute drag force damping utilizing the DampingMM solver within the Coventorware software. Table 5 compares numerical and experimental quality factors associated with drag force damping for the first transverse oscillation

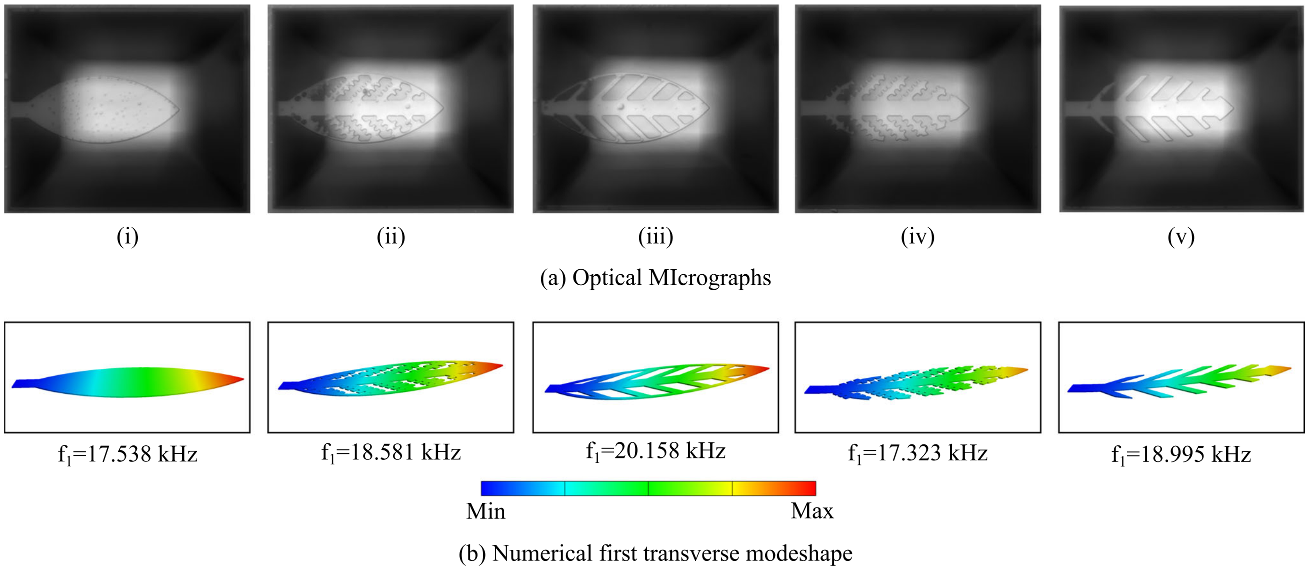



**Fig. 10** **a** Optical micrographs of fabricated leaf-shaped micro-cantilever beams, namely (i) Leaf 1,(ii) Leaf 2,(iii) Leaf 3,(iv) Leaf 4, and (v) Leaf 5, **b** the corresponding numerically computed first transverse mode shape (mode 1)

**Table 3** Comaprision of numerical and experimental first transverse mode natural frequency ($f_1$) of leaf-shaped micro-cantilever beams

| Conf. | Mode 1 ($f_1$) in kHz | | | | |
|---|---|---|---|---|---|
| | Leaf 1 | Leaf 2 | Leaf 3 | Leaf 4 | Leaf 5 |
| Num | 17.538 | 18.581 | 20.158 | 17.323 | 18.995 |
| Exp | 18.252 | 18.131 | 18.133 | 17.772 | 18.426 |
| Error (%) | 3.911 | 2.481 | 11.167 | 2.526 | 3.088 |

**Table 4** Comparison between numerical and experimental natural frequencies of the second transverse and torsional modes for various leaf-shaped microcantilever beams

| Conf. | Mode 2 ($f_2$) in kHz | | | Mode 3 ($f_3$) in kHz |
|---|---|---|---|---|
| | Num. | Exp. | Error (%) | Num. |
| Leaf 1 | 141.261 | 132.654 | 6.488 | 103.795 |
| Leaf 2 | 121.752 | 121.353 | 0.328 | 99.841 |
| Leaf 3 | 130.682 | 127.324 | 2.637 | 109.669 |
| Leaf 4 | 99.087 | 104.744 | 5.4 | 106.653 |
| Leaf 5 | 111.234 | 115.314 | 3.538 | 134.319 |

mode. The numerical quality factors range from $10^2$ to $10^1$, whereas the experimental quality factors consistently fall within the $10^1$ range. The difference in quality factor values arises from the use of linear models in the DampingMM solver, which tends to estimate higher quality factor values. This discrepancy highlights the limitations of linear modeling in accurately predicting damping characteristics. Based on the experimental and numerical findings, it can be concluded that drag force damping is the primary energy dissipation mechanism in leaf-shaped microcantilever beams. This implies that the interaction between the beam and the surrounding fluid or medium, which generates drag forces, is crucial in how these beams lose energy. Further research could explore ways to minimize the drag force damping through changes in beam geometry and environmental conditions. A comprehensive comparison of the present model with various geometries documented in the literature is provided in Appendix II of the supplementary material. This section details the differences and advantages of our novel design, highlighting how it stands in relation to existing configurations.

**Table 5** Comaprision of numerical drag and experimental quality factor (Q) associated with first transverse oscillation mode for different leaf-shaped micro-cantilever beams

| Conf. | | Leaf 1 | Leaf 2 | Leaf 3 | Leaf 4 | Leaf 5 |
|---|---|---|---|---|---|---|
| Quality Factor (Q) | Num | 106.119 | 89.609 | 84.519 | 75.527 | 71.158 |
| | Exp | 24.129 | 32.571 | 35.752 | 18.375 | 16.883 |

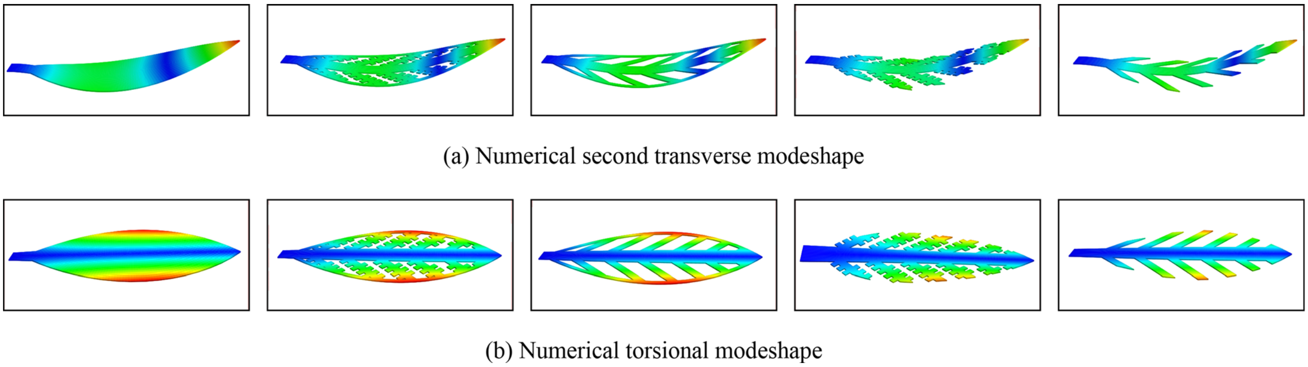



**Fig. 11** **a** Numerically computed second transverse mode shape (mode 2), and **b** numerically computed torsional mode shape (mode 3)

## 5.7 Anchor loss damping

Classical loss mechanisms (drag and squeeze film damping) in oscillating leaf shaped beams are thoroughly discussed in the previous section. Another significant loss mechanism in MEMS devices is commonly termed anchor loss, also referred to as anchor damping, support loss, clamping loss, or attachment loss. This occurs when a portion of the elastic energy generated during the vibration of the device propagates into the surrounding structure to which it is attached. previous studies, including those by David (2005) and John et al. (2007), often assume that any energy transmitted through the anchors is irretrievably lost. From the structural perspective in general, this energy dissipation is equivalent to losses caused by classical damping mechanisms.

Figure 12a illustrates the modeling of anchor loss damping using the Coventorware (Source yyy) numerical tool. To prevent elastic waves from reflecting back into the device, a Perfectly Matched Layer (PML), also known as a Quiet Boundary, is applied to the outer boundary of the substrate in the model. In the modeling process, the dimensions of the Quiet Boundary are carefully selected based on the acoustic wavelength, ($\lambda$), to ensure accurate simulation of wave propagation and absorption. The acoustic wavelength is defined as:

$$\lambda = \frac{1}{f}\sqrt{\frac{E}{\rho}} \tag{4}$$

where $f$ represents the natural frequency of the system, $E$ is the Young's modulus of the material, and $\rho$ denotes the density of the leaf-shaped microcantilever beams. Using the MemMech solver in Coventorware, the dynamic response of the leaf-shaped microcantilever beams has been simulated and is presented in Fig. 12b. From the results depicted in Fig. 12b, the inverse quality factor associated with anchor loss, $1/Q_{AL}$, for the first transverse mode of Leaf 1 is calculated to be $7.72688 \times 10^{-12}$. This value highlights the effectiveness of the modeling approach

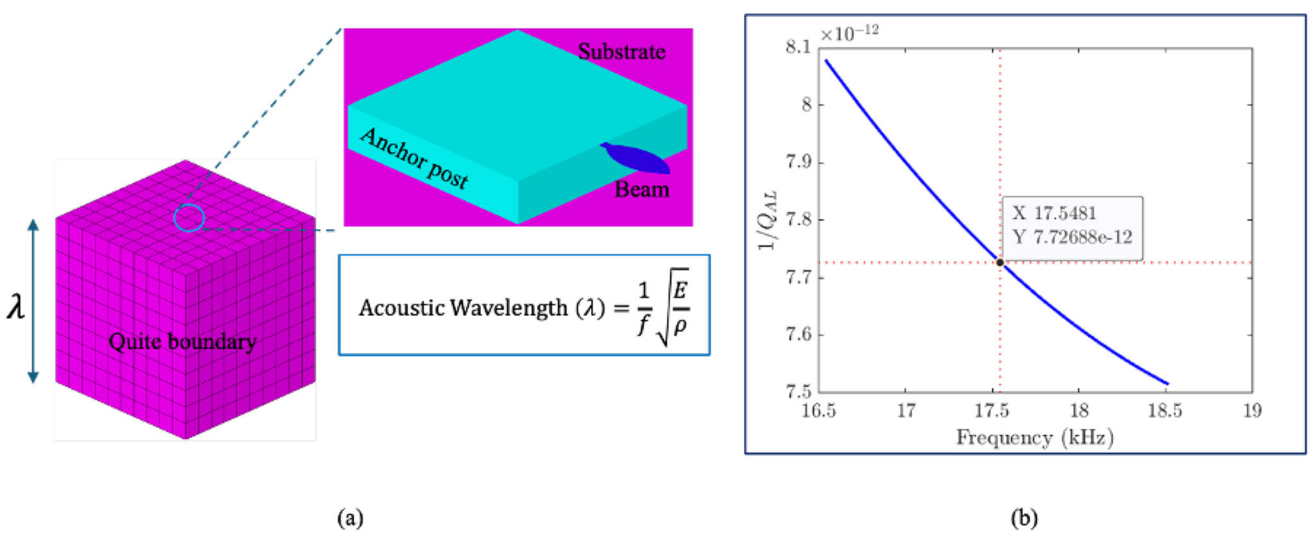



**Fig. 12** **a** Modeling of Anchor loss damping mechanism in Coventorware, **b** variation in anchor loss quality factor with respect to the excitation frequency

**Table 6** Quality factor ($Q_{AL}$) associated with anchor loss damping for the first transverse oscillation mode for different leaf-shaped micro-cantilever beams

| Conf. | Leaf 1 | Leaf 2 | Leaf 3 | Leaf 4 | Leaf 5 |
|---|---|---|---|---|---|
| $Q_{AL}$ | $1.2932 \times 10^{11}$ | $1.4397 \times 10^{11}$ | $1.3835 \times 10^{11}$ | $2.1053 \times 10^{11}$ | $2.0554 \times 10^{11}$ |

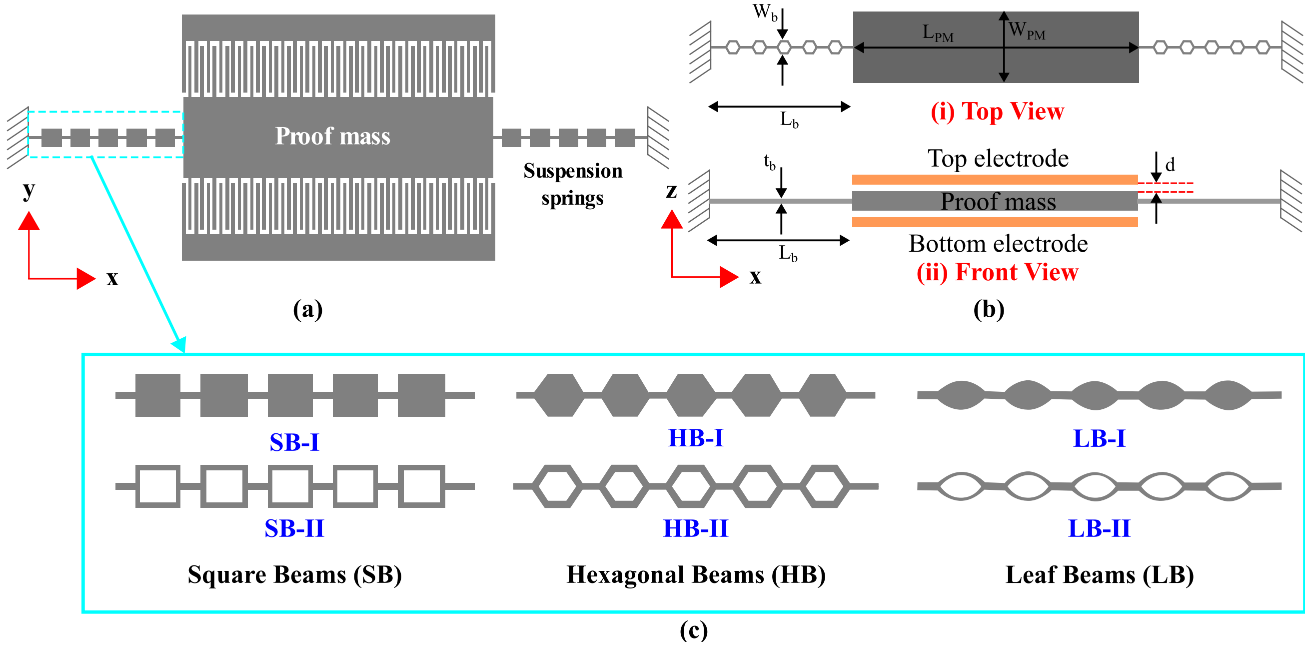



**Fig. 13** **a** Schematics of MEMS accelerometer for in-plane sensing; **b** Schematics of MEMS accelerometer for out-of-plane sensing; **c** different suspensions springs

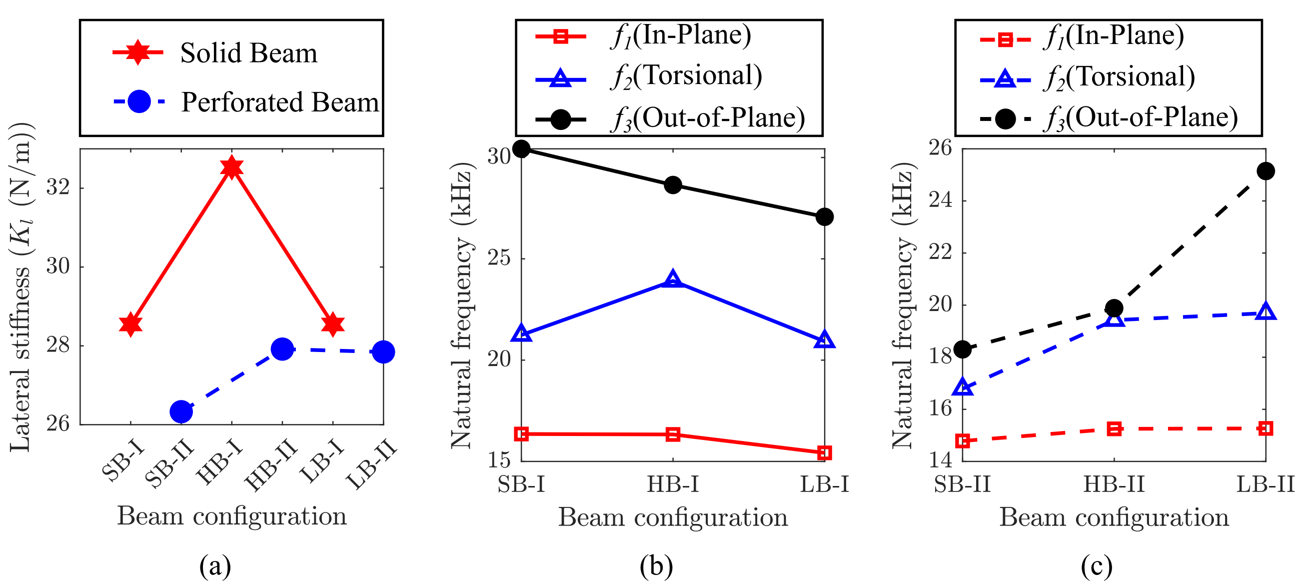



**Fig. 14** Numerical values of **a** lateral stiffness ($K_l$ in N/m); **b** First three natural frequencies ($f_1$ (in-plane mode),$f_2$ (torsional mode), and $f_1$ (out-of-plane mode)) of MEMS accelerometer with different solid beam configurations such as square (SB-I), hexagonal (HB-I), and leaf (LB-I) beams; **c**

in capturing the anchor loss characteristics of the microcantilever beams and serves as a critical parameter in evaluating the damping behavior and performance of the MEMS microcantilever beam.

The computed $Q_{AL}$ values for various leaf-shaped microcantilever beams are summarized in Table 6. From the results, it is evident that the order of $Q_{AL}$ for anchor loss damping is approximately $10^{11}$. In contrast, for fluid damping, the Q-factor is significantly lower, typically ranging from $10^1$ to $10^2$. This comparison highlights the relative influence of different damping mechanisms under varying environmental conditions. Anchor loss damping plays a prominent role under vacuum conditions due to the absence of fluid-induced energy dissipation. Conversely, in ambient conditions, fluid damping becomes the dominant energy dissipation mechanism, overshadowing the effects of anchor loss.

### 5.8 Application in MEMS accelerometer

Subsequently, We utilized various beam geometries as suspension springs in a MEMS accelerometer, as illustrated in Fig. 13a, to investigate their influence on the performance. The geometries included square beams (SB), hexagonal beams (HB), and leaf-shaped beams (LB), designed in both solid and perforated forms. Each suspension spring measured 300 $\mu m$ in length and 35 $\mu m$ in width. Numerical simulations were performed to calculate the first three natural frequencies, shown in Fig. 14b, c. Static displacement analysis revealed that perforated beams exhibited higher flexibility than solid beams, as depicted in Fig. 14a. Among the solid beams, hexagonal beams (HB-I) had the highest stiffness ($K_l$), with stiffness values increasing from SB to HB and then decreasing from HB to LB. The first three mode natural frequencies: in-plane ($f_1$), torsional ($f_2$), and out-of-plane ($f_3$)of solid beams decreased as the shape transitioned from SB to LB, as shown in Fig. 14b. In contrast, for perforated beams, the natural frequencies increased as the shape changed from SB to LB, illustrated in Fig. 14c. The advantage of using leaf-shaped beams in MEMS accelerometers is their ability to effectively separate different vibration modes. This separation enhances the device's performance and accuracy, providing a distinct superiority over other beam shapes, such as square or hexagonal beams. The unique geometry of leaf-shaped beams allows for better isolation of in-plane, torsional, and out-of-plane modes, resulting in improved frequency tuning and overall device functionality. Thus, employing different-shaped beams, both with and without perforations, provides an effective method for tuning the frequency of MEMS accelerometers to meet specific application requirements.

In addition to the modal and stiffness analyses, we conducted a static displacement analysis of the in-plane mode MEMS accelerometer to evaluate its performance under an input acceleration of $\pm30g$ in y-direction. This analysis aimed to determine the maximum displacement experienced by the proof mass (PM) for various beam configurations. The results of this analysis are presented in Fig. 15a. The analysis revealed the following order of maximum displacement for the different beam configurations: SB-I $<$ HB-I $<$ LB-I $<$ LB-II $<$ HB-II $<$ SB-II. This

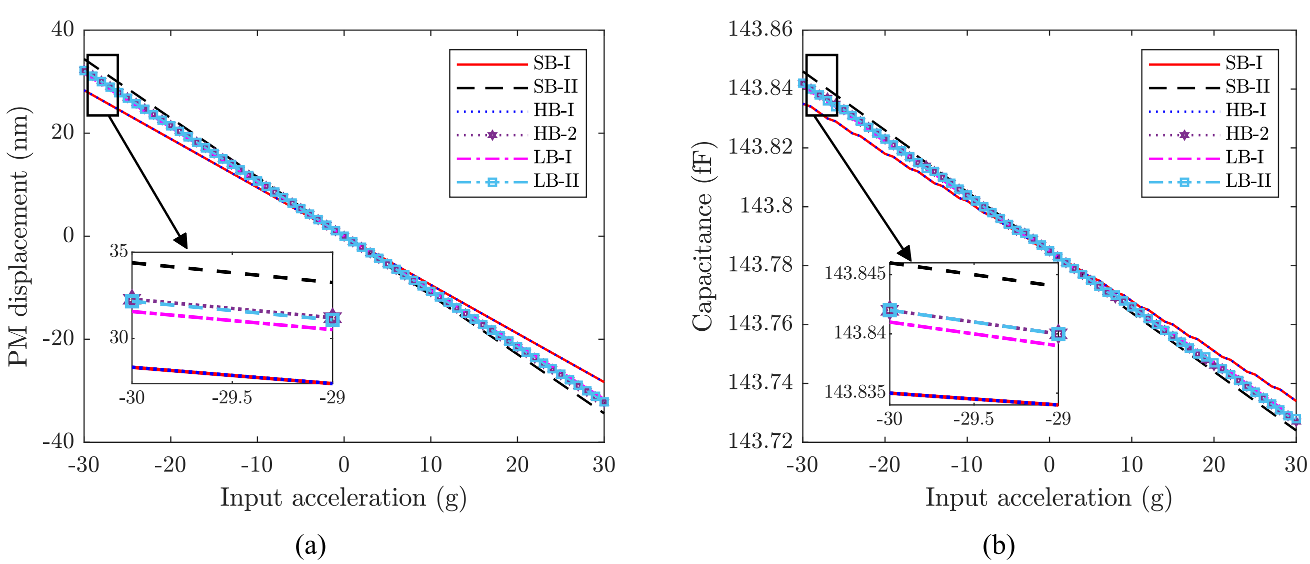



**Fig. 15** **a** Displacement vs acceleration; **b** Capacitance vs acceleration **c** maximum displacement experienced by the proof mass, and **d** sense capacitance due to change in the overlap area of comb figure of MEMS accelerometer with different suspension springs under the influence of input acceleration of $\pm30g$, in the y-direction. Here $g$ is the acceleration due to gravity of 9.81 $m/s^2$

ordering indicates that the solid square beam (SB-I) experiences the least displacement, while the perforated square beam (SB-II) undergoes the higher displacement. The configurations HB-I, LB-I, LB-II, and HB-II exhibit displacement values between those of SB-I and SB-II. As solid beams, HB-I and LB-I show relatively lower displacements, indicating higher stiffness than LB-II and HB-II, which have perforations. The results highlight a clear trade-off between stiffness and flexibility. Solid beams offer minimal displacement and higher stiffness, which are desirable for high-precision applications where structural rigidity is crucial. Conversely, beams with perforations, while providing increased flexibility and potentially beneficial damping properties, result in higher displacements, which may affect the accuracy and sensitivity of the accelerometer. Subsequent to the static displacement analysis, we examined the sense capacitance formed between the comb fingers due to the change in the overlap area. The sense capacitance is a crucial parameter that reflects the sensitivity of the MEMS accelerometer to input accelerations. The results of this analysis are depicted in Fig. 15b. When the accelerometer is at rest, with zero input acceleration, the nominal or base capacitance was measured to be 143.78 fF. The sense capacitance varies in response to the displacement of the proof mass, which changes the overlap area between the comb fingers. Consistent with the displacement results, the sense capacitance for the MEMS accelerometer with the perforated square beam (SB-II) configuration is higher compared to other beam configurations. The findings highlight a trade-off between sensitivity and structural rigidity. Beams with higher sense capacitance (like SB-II) offer greater sensitivity to acceleration but may be more prone to inaccuracies due to higher flexibility and potential for noise. Solid beams (like SB-I) provide lower sense capacitance changes, which might reduce sensitivity but offer improved accuracy and stability.

Finally, we extended our analysis to the out-of-plane (z-axis) MEMS accelerometer, depicted in Fig. 13b. The out-of-plane MEMS accelerometer features a pair of electrodes positioned on the top and bottom of the proof mass. These electrodes are designed to form a parallel-plate capacitor, where the relative displacement between the electrodes and the proof mass influences the capacitance. This device employs a differential capacitance ($\Delta C = C_2 - C_1$) mechanism, utilizing top and bottom electrodes that are separated by a gap, 'd' of 5 $\mu m$. Here, $C_1$ is the capacitance between the bottom electrode and the proof mass, whereas $C_2$ is the capacitance between the top electrode and the proof mass. When an acceleration is applied along the z-axis, the proof mass moves, changing the distance between the top and bottom electrodes. This displacement causes a change in the gap, leading to a variation in the differential capacitance.

Figure 16a, b illustrate the maximum displacement experienced by the proof mass and the corresponding change in capacitance ($\Delta C$) for an out-of-plane mode MEMS accelerometer subjected to an input acceleration of $\pm 30g$ in the z-direction. The analysis focuses on evaluating different suspension spring configurations to determine their impact on the performance of the accelerometer. The analysis reveals significant variations in the maximum displacement of the proof mass across different beam configurations. The MEMS accelerometer with the

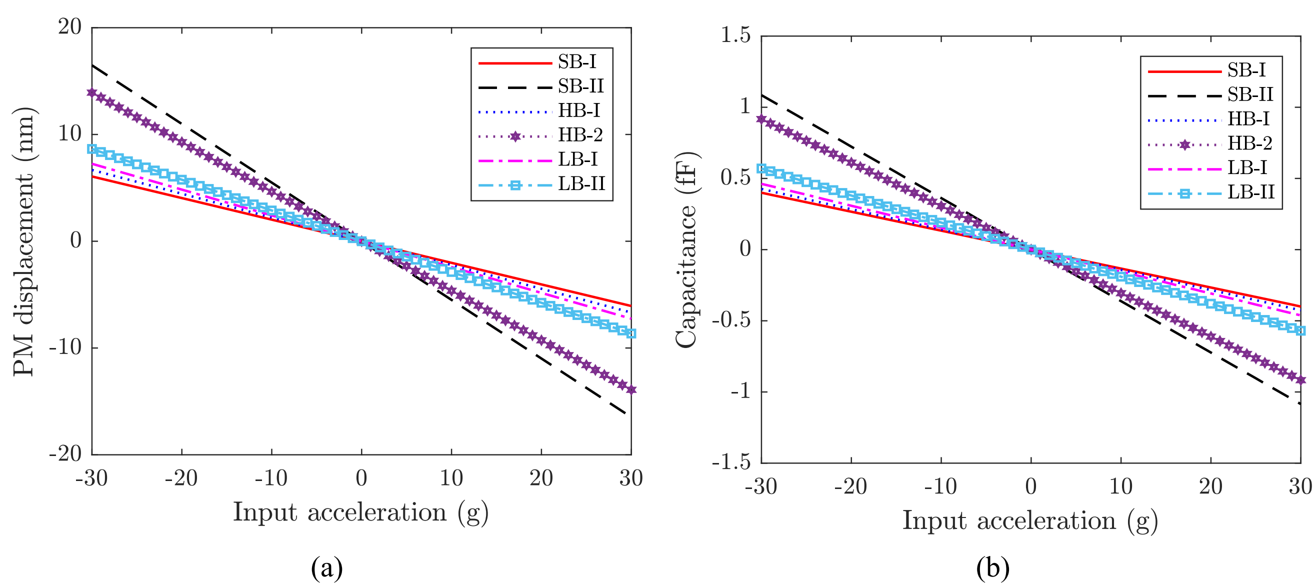



**Fig. 16** **a** Maximum displacement experienced by the proof mass, and **d** sense capacitance due to change in a gap of comb figure of MEMS accelerometer with different suspension springs under the influence of input acceleration of $\pm 30g$, in z-direction

perforated square beam (SB-II) shows the highest displacement. In contrast, the solid square beam (SB-I) configuration exhibits the lowest displacement. The variation in capacitance corresponds to the displacement of the proof mass. A larger displacement leads to a more significant change in capacitance. Similar to the displacement results, the SB-II configuration shows the highest $\Delta C$, while the SB-I configuration shows the lowest. The comprehensive analysis of the out-of-plane MEMS accelerometer has highlighted significant insights into the performance implications of various suspension spring configurations. Our findings reveal that the choice of beam configuration plays a crucial role in determining the performance of MEMS accelerometers for better functionality. Overall, our study provides the importance of carefully selecting and designing suspension springs in MEMS accelerometers to achieve the desired balance of sensitivity, accuracy, and stability. Leaf-shaped suspension springs can offer significant advantages for MEMS accelerometers across various industrial applications due to their enhanced sensitivity, precision, durability, and energy efficiency. These beams enable improved frequency tuning and mode separation, making them ideal for precision sensing tasks such as vibration monitoring, tilt sensing in electronics, and navigation systems. The distinct in-plane, torsional, and out-of-plane modes reduce cross-talk, enhancing measurement accuracy and device reliability. Their durability allows them to withstand high accelerations (up to $\pm 30$ g), which is essential for automotive and aerospace applications, including airbags deployment, stability control, and flight instruments. Additionally, the optimized damping and high quality factors of leaf-shaped beams lower power consumption, a crucial feature for power efficient devices like fitness trackers and smartphones, extending their usability. This study acknowledges the practical relevance of leaf-shaped beams in MEMS accelerometer technology, especially for automotive, aerospace, electronics, and healthcare industries. Future research will focus on fabricating the designed MEMS accelerometer to allow empirical testing and validation of these benefits, confirming the real-world applicability and performance.

## 6 Conclusions

This study explored the frequency and damping characteristics of Microelectromechanical Systems (MEMS) cantilever beams with various shapes, including uniform, diamond, hexagonal, and leaf-shaped configurations. These shapes were derived from a stepped rectangular cantilever beam by sweeping the vertical faces, and their characteristics were determined through comprehensive numerical simulations. Additionally, experiments were conducted specifically on leaf-shaped beams to validate the simulation results, showing good agreement with a maximum percentage error of 11.2%. Further, Our findings highlight the significant impact of beam geometry on the performance of MEMS accelerometers. Specifically, we demonstrated that leaf-shaped beams offer superior flexibility for tuning the frequency to meet specific application requirements. The unique geometry of leaf-shaped beams allows for the effective separation of different vibration modes, such as in-plane, torsional, and out-of-plane, enhancing the overall performance and accuracy of the MEMS accelerometers. This mode separation provides a distinct advantage over other beam shapes, such as square or hexagonal beams, by improving frequency tuning and isolating different vibrational modes more efficiently. Furthermore, it is seen that employing various shaped beams, both with and without perforations, provides an effective method for fine-tuning the frequency characteristics of MEMS accelerometers. Perforated beams, in particular, exhibited higher flexibility compared to solid beams, further expanding the design possibilities for optimizing device performance.

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**Data availability** All relevant data supporting the findings of this study have been included in the manuscript.

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